



A remote phosphor film of silicate-poly(styrene-co-glycidyl methacrylate) composites for NUV chip-based white LED



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ABSTRACT

Herein we adopt a simple route to prepare silicate phosphor-containing transparent composite film which is able to emit white light on the base of a blue emitting co-polymer matrix poly(styrene-co-glycidyl methacrylate) under excitation of 365 nm. SEM micrographs indicate that the surface treated silicate phosphor particles modified by silane coupling agent and embedded in co-polymer matrix separate from each other with a certain spacing and without agglomeration. More importantly, mutual enhancement in characteristic emission peak of two ingredients is observed. By varying the weigh percentage of silicate phosphor, yellow-to-white photoluminescent color-tuning could be readily accomplished depending on the variation of the intensity ratios between the characteristic yellow emission of silicate phosphor and the blue luminescence of the polymer matrix. And white emission [CIE: (0.354,0.322)] could be obtained from the simultaneous blue and yellow emission (intensity ratios value is 0.93). Simple NUV-based WLEDs were fabricated using prepared silicate phosphor doped composite material, showing T_c of 5917 K, CRI of 84.6. When a larger size composite film was simply assembled with COB, large area white lighting device with a certain color reduction came into being. All the results indicate that the of silicate phosphor-containing transparent composite film has practical significance for the packaging of remote phosphor converted white LEDs.

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1. Introduction

Phosphor-converted white light-emitting diodes (pc-WLEDs) have undergone rapid development as an indispensable solid state light source for the next generation lighting industry and display systems owing to their superior advantages such as long lifetime, energy saving, high efficiency, low weight and environmental friendliness [1,2]. For this approach, the convenient way is the “blue

chip + yellow phosphor” mode, the combination of blue and yellow light makes a white-emitting LED. For further minishing the heating of the phosphor-containing active layer from the LED chips, some efforts have been focused on remote LED encapsulation [3], among which the most common is placing polymeric composite film away from the chip die. [4,5] To this end, polycarbonate (PC) [6], polyethylene glycol terephthalate (PET) [7], polymethylmethacrylate (PMMA) [8], and epoxy resins [9] are some of the most used polymer matrix. However, this mode suffers not only the proportional imbalance of blue and yellow light but the color drifting also, because blue light, as a primary color emitted by the chip, is strongly influenced by temperature and driving current [10,11]. Satisfyingly, UV-based phosphor converted WLED can conquer the aforementioned pitfalls owing to the invisible high-energy emission of the LED chip. Thus, the current focus of the fabrication shifts gradually from blue chip + yellow phosphor to ultraviolet chip + RGB tricolor phosphor. Nevertheless, there still

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exist some problems for WLEDs fabricated with ultraviolet-LED chips and tri-color phosphors. Generally, the luminescent efficiency is relatively low in this system owing to the strong reabsorption [12] of blue light by red and green phosphors as well as mutual excitation [13] among luminescent center ions (like Eu^{3+} and Tb^{3+}). Besides, the tuning of the three primary colors needs heaps of times of experiments using conditional trial-and-error method in RGB mode [14]. At the same time, “UV chip + Dy(III)-MOFs” mode has been touched [15–17]. White-light emission can be achieved through a simpler BY dichromatic strategy on the balance of blue (B) and yellow (Y) colors from Dy(III)-MOF. In contrast to the “UV-chip + RGB primary phosphors” strategy to utilize mixed-metal (e.g. Eu^{3+} , Tb^{3+} and Zn^{2+}) for white-light emission, this provides an alternative way to obtain white-light phosphors with even easier preparation and manipulation even easier. However, it is worth noting that Dy^{3+} can not be effectively excited by ultraviolet because its charge transfer band and 5d energy level both locate above 5000 cm^{-1} . In addition, the f-f transitions in the near ultraviolet region are forbidden transitions and the corresponding luminescence efficiency is low [18]. Therefore, it is an urgent but promising task to propose a multifunctional polymer being different from above mentioned ones (PET, PMMA, PC and epoxy resins) as a matrix for phosphor particles to realize good film formation property and act as blue emitting luminophors for the combination with yellow emission applied in NUV-based LEDs. The relevant researches have been rarely reported to the best of our knowledge.

Hence, we propose a new polymer matrix poly(styrene-co-glycidyl methacrylate) (noted as PS-PGMA). Subsequently, a well dispersed silicate phosphor particle transparent composite film was prepared by using a readily accessible melt compounding process, in which the co-polymer matrix PS-PGMA plays dual functions. On one hand, it enables the easiness of film formation and low cost of massive production; on the other hand, it is strongly blue luminescent with its blue emission in the system being sufficient to balance yellow emission of silicate phosphor to generate white-light. Moreover, by varying the ratios of silicate phosphor to PS-co-PGMA, yellow-to-white PL color-tuning can be readily accomplished depending on the variation of the intensity ratios between the characteristic yellow emission and the blue luminescence. White emission [CIE: (0.354,0.322)] could be obtained from blue emission and comparable yellow emission (intensity ratios value is 0.93). Simple NUV-based WLEDs were fabricated using prepared composite film, showing T_c of 5917 K, CRI of 84.6. Additionally, when a larger size composite film was simply assembled with COB, large area white lighting device with a certain color reduction came into being. All the results indicate that this kind of silicate phosphor-containing transparent composite film has practical significance for the packaging of remote phosphor converted white LEDs.

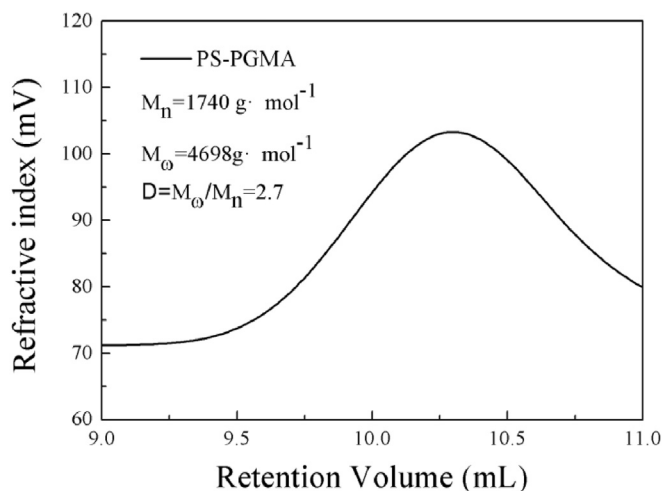


Fig. 1. The GPC chromatogram of co-polymer matrix.

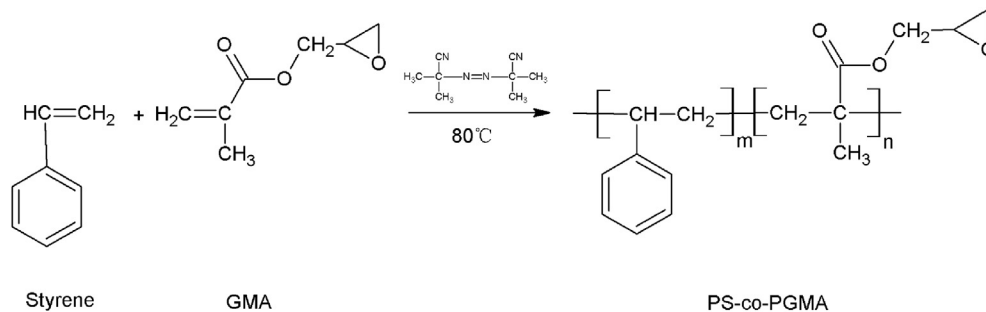
2. Experimental

2.1. Materials

Polymer matrix PS-PGMA was synthesized by the copolymerization of styrene and glycidyl methacrylate (purchased from Alfa Aesar) in the previous experimental work of our group, and the synthesis route is depicted in Scheme 1. According to the GPC measurement of PS-PGMA in Fig. 1, weight average molecular weight M_w is 4698 g/mol and number average molecular weight M_n is 1740 g/mol. Accordingly, the polydispersity indexes (PDI) is calculated as 2.7 by M_w/M_n . Orange emitting phosphor (Silicate: Eu^{3+}) was purchased from Alfa Aesar. As shown in Fig. 2(b), the silicate phosphor particles haven't a peculiar morphology, among which the size of particles is mainly polydispersed in 10–20 μm . The inset is the histogram of silicate phosphor particle size distribution (after counting 85 particles) in order to visualize the polydispersion. Coupling reagent N-[3-(Trimethoxysilyl)propyl]ethylenediamine (Z-6020) and surfactant hexadecyl trimethyl ammonium bromide (CTMAB) were purchased from Dow Corning. Ultraviolet LED chip as well as COB LED and PTFE coagulating molds were used in this work.

2.2. Apparatus

The molecular weight of polymer matrix PS-PGMA was determined using gel permeation chromatography (GPC, TDAmex, Malvern). UV–vis absorption and transmittance spectra were measured on a Hitachi-3900 UV-VIS 5 spectrometer (the slit width



Scheme 1. Synthesis route of polymer matrix PS-PGMA.

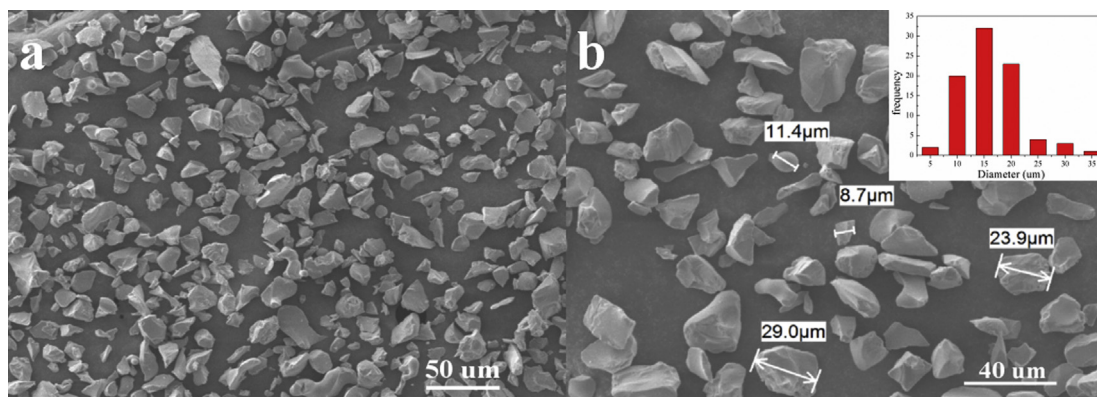


Fig. 2. SEM micrograph of silicate phosphor, the inset profile is the histogram of silicate phosphor particle size distribution (after counting 85 particles).

is 2 nm). Fluorescence spectra were recorded on an Edinburgh LFS-920 spectrometer (the slit width is 1 nm). Electroluminescence (EL) measurements were conducted on a computer controlled PMS-80 UV–vis–near IR spectrometer with an integrating sphere. Thermogravimetric/differential thermal analysis was performed on an STA409C Analyzer. Fluorescence lifetime was recorded using an FLS920 combined steady state and lifetime spectrometer. The morphologies of raw silicate phosphor and remote phosphor film were characterized using field emission scanning electron microscope (FE-SEM, JSM-6700F). The surface topography of the polymer phosphor film was measured by atomic force microscope (AFM, SPA-300 HV). All measurements were made at room temperature unless otherwise stated.

2.3. Synthesis of polymer matrix PS-PGMA

A tetrahydrofuran solution containing styrene, GMA and 2,2'-Azobisisobutyronitrile (AIBN) was transferred to a four-necked flask and mixed under stirring. After dissolution, the solution was degassed with N₂ for 30 min to exclude the O₂ dissolved in the solution. Then, the flask containing the reaction solution was sealed and heated at predetermined temperature for 10 h in a water bath. After polymerization, the resulting solution was precipitated in methanol and filtered off. The crude reaction product was further purified by several reprecipitations from acetone to methanol three times. Lastly, the copolymers were dried under vacuum at 50 °C to obtain white solid powder.

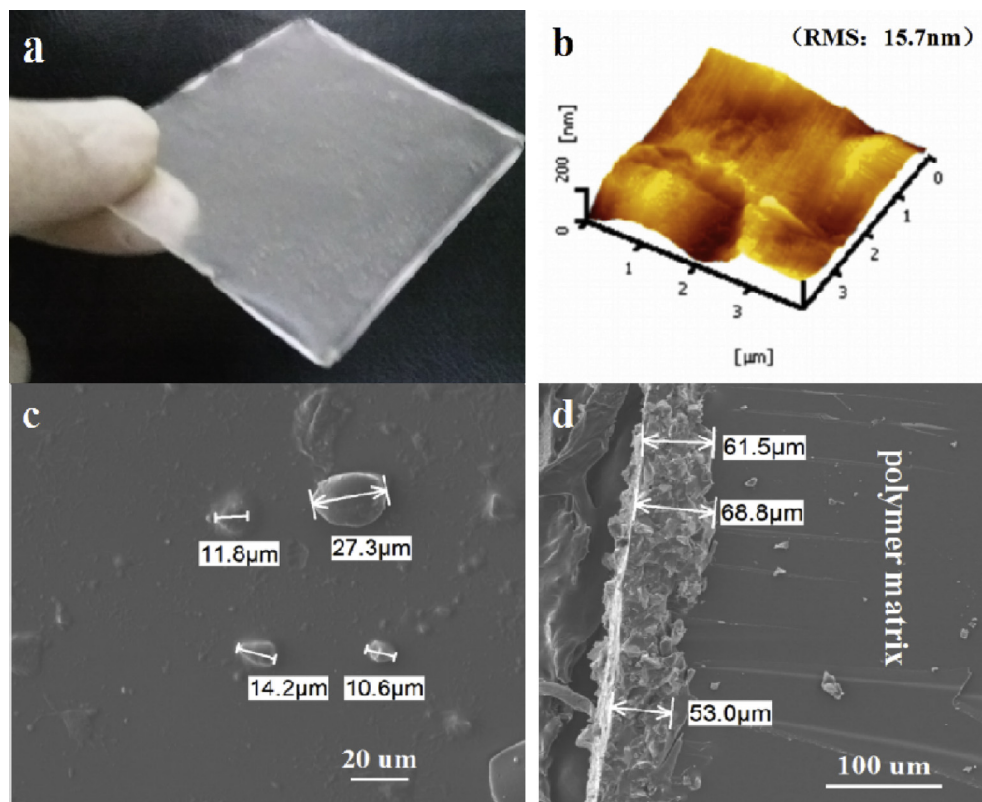


Fig. 3. Morphologies of silicate phosphor polymeric composite film (a) appearance, (b) AFM micrograph, (c) and (d) are the SEM micrographs.

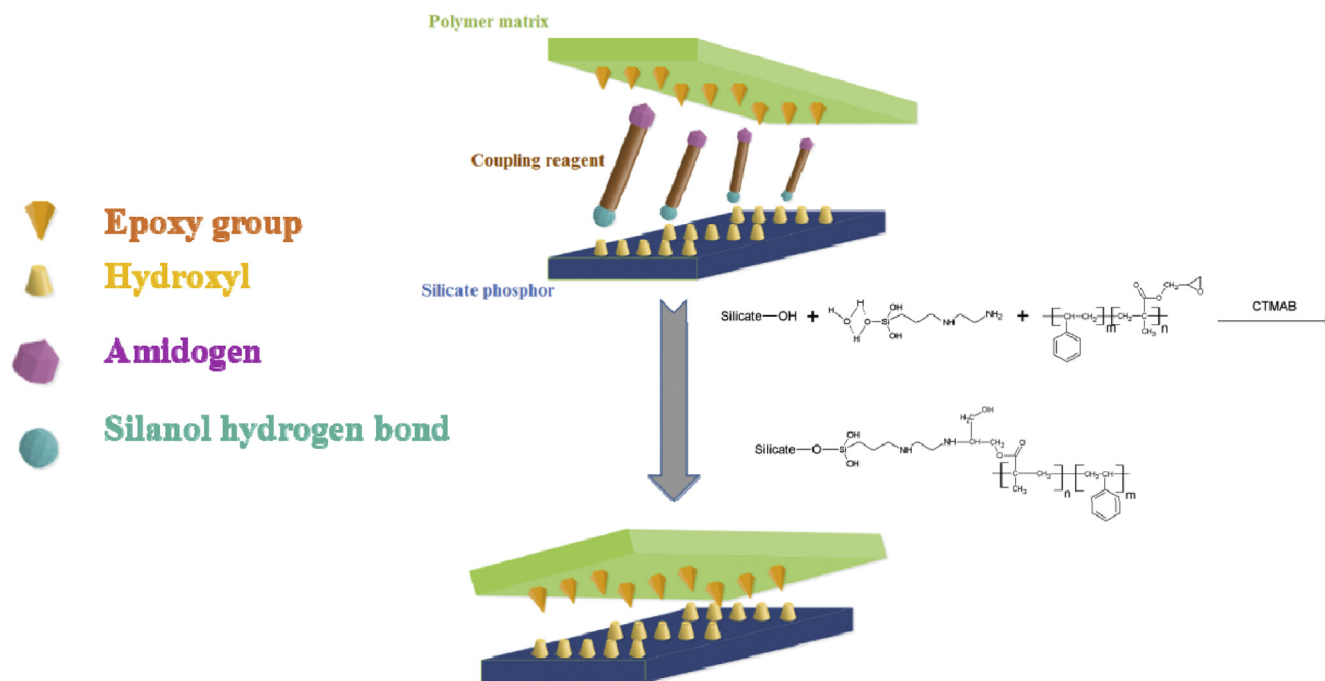


Fig. 4. The action mechanism of silane coupling agent.

2.4. Surface treatment of silicate phosphor

The flask with 50 ml of ethanol was heated up to 70 °C, surfactant CTMAB was added under stirring. Five minutes later, silicate phosphor was added. Another 5 min later, the coupling agent Z-6020 was slowly dripped into the flask under stirring for 15–20 min. At last, the phosphor was separated from the liquid.

2.5. Preparation of the silicate phosphor composites film

The polymer matrix PS-PGMA and surface treated silicate phosphor were dried in an oven at 60 °C for 8 h before use. Polymer matrix in fine powder was added into a 50 ml crucible and first melted for 3 min at 180 °C. Then, different amounts (0.1, 0.3, 0.5, 0.7 and 1 wt.%) of the silicate phosphor were added into the molten polymer and mixed for a further 5 min before the mixture became completely melted. The samples were then melt pressed into a PTFE recipient under this temperature. 20 min later, the recipient was left to room temperature for solidification. Pure PS-PGMA, as a control sample, was prepared following the same procedure.

3. Results and discussion

3.1. Micro-morphology of the silicate phosphor polymeric composite film

Fig. 3(a) exhibits the appearance of the silicate phosphor composite transparent polymer film prepared by the process proposed above. Fig. 3(b) shows the surface topography of polymer phosphor film side contacted with the surface of mold, which is relatively flat with a surface roughness RMS of 15.7 nm. In Fig. 3(c), silicate phosphor particles are embedded in PS-PGMA and separated from each other with a certain spacing rather than being reunited. In this system, silane coupling agent plays a role as a bridge in promoting the dispersion between silicate particles as well as the compatibility between inorganic silicate phosphor and organic polymer.

The acting mechanism of silane coupling agent is simply shown in Fig. 4 and the IR spectra of related materials are depicted in Fig. 5 to check the hypothesis. For the contrast between (1) surface treated silicate phosphor and (2) raw silicate phosphor, the reaction between silicate phosphor and coupling agent can be confirmed by

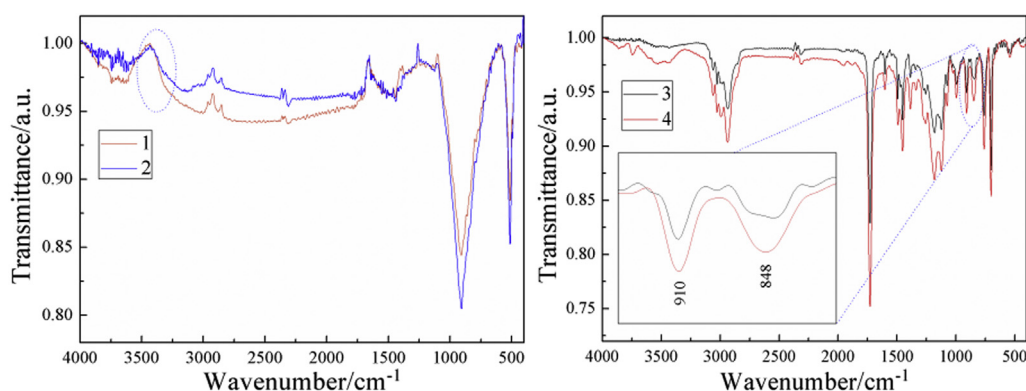


Fig. 5. IR spectra of surface treated silicate phosphor (1), raw silicate phosphor (2), PS-PGMA/Si after melt blending (3) and PS-PGMA (4) in the solid state.

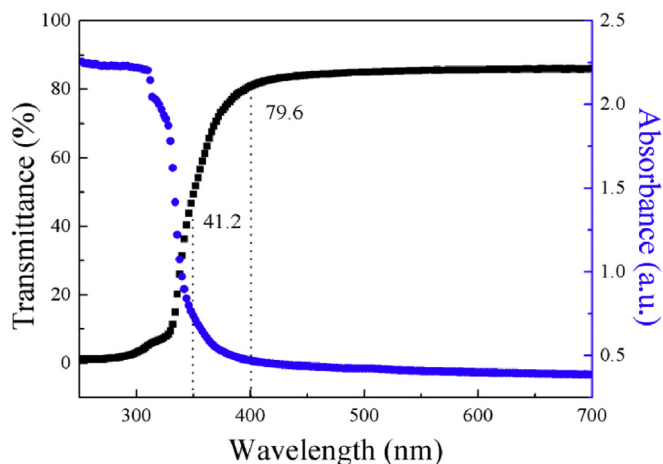


Fig. 6. Light transmittance (square) and absorption spectrum (circle) of composite polymeric film in UV–visible wavelength.

the broadened band at 3500 cm^{-1} . Furthermore, from the contrast between (3) PS-PGMA/Si after melt blending and (4) PS-PGMA, we can see the bands at 910 cm^{-1} and 848 cm^{-1} ascribed to epoxy group becomes broaden, which verifying the reaction between epoxy group in PS-PGMA and amidogen in coupling agent. This is a welcome phenomenon since it is well known that the luminescent intensity is related to the average distance between luminescent centers and agglomeration would lead to a decreased luminous performance of the hybrid membrane. It is noted that larger particles tend to be completely exposed on the substrate surface, and smaller particles only partially stick out the surface of matrix. This

is because silicate phosphor particles gradually subside into one side of polymer matrix in the completely molten state during the mixing process because of gravity and immiscibility. According to the cross-sectional SEM image of the phosphor layer, the thickness of the silicate phosphor which accumulate compactly in PS-PGMA gives a mean value of $61.1\text{ }\mu\text{m}$ (Fig. 3(d)).

3.2. Optical properties

Refer to Fig. 6, the transmission spectrum of the silicate phosphor composite film exhibits $>80\%$ transmittance in the region $400\text{--}700\text{ nm}$, with transmittance gradually declining to a value of 41.2% recorded at the wavelength of 365 nm . This satisfies the requirement for the remote phosphor film applied in LEDs, with a higher transmittance of emitted light in the visible spectral range and an efficient absorption of excitation in the UV region.

The excitation and emission spectra of silicate phosphor and polymer matrix film in solid state recorded at room temperature are shown in Fig. 7. The excitation spectrum of silicate phosphor, as can be seen in Fig. 7(a), obtained by monitoring the emission wavelength at 580 nm covers the wavelength range from 240 to 500 nm with a maximum at 364 nm , Which is attributed to the $4f \rightarrow 5d$ transition of europium ions [19]. Upon the excitation with a wavelength of 365 nm , the emission spectrum exhibits the characteristic emission at 580 nm attributed to the allowed $4f^6 5d^1 \rightarrow 4f^7$ transition of europium ion [20]. For excitation spectra of polymer matrix film when monitored at the wavelength of 420 nm shown in Fig. 7(b), two discriminable bands centered at 307 and 369 nm appear, which are ascribed to the $\pi\text{--}\pi^*$ transition of benzene ring [21] in polystyrene segment and the $n\text{--}\pi^*$ transition of carbonyl [22] in poly(glyceryl methacrylate) segment, respectively. Excited by the wavelength of 365 nm , co-polymer matrix exhibits in its

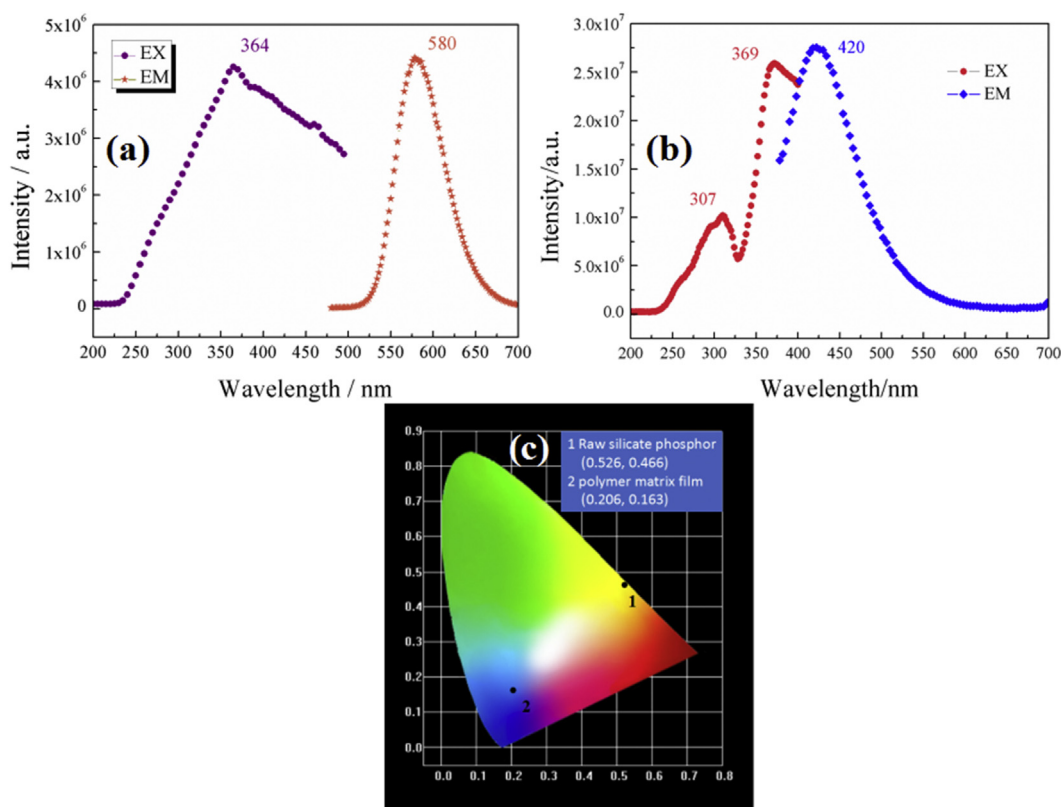


Fig. 7. PLE and PL spectra of raw silicate phosphor (a), polymer matrix film (b) and the CIE chromaticity coordinates (c) raw silicate phosphor (1) and polymer matrix film (2).

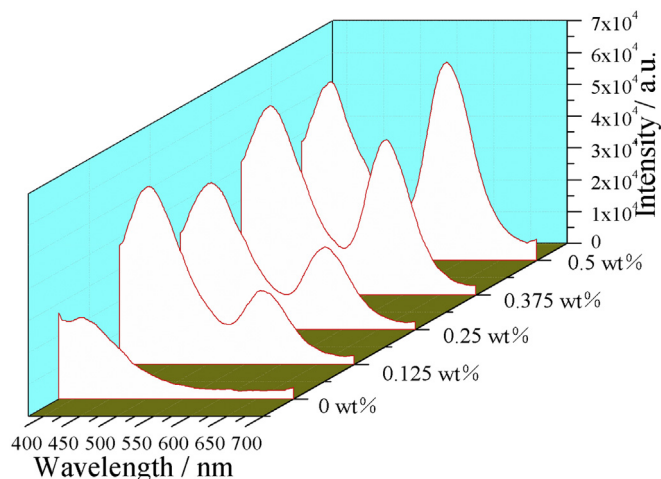


Fig. 8. Fluorescence emission spectra excited at 365 nm of serial composite phosphors (0 wt%–0.5 wt%).

emission spectrum a broad band centered at 420 nm. The corresponding CIE chromaticity coordinates are calculated from the emission spectra and marked in Fig. 7(c). It can be observed that the CIE chromaticity coordinates of the silicate phosphor and the copolymer matrix respectively locate in orange yellow and blue light zone. In view of this, it is a feasible proposal to prepare a white light emitting silicate phosphor composite film by taking advantage of the bi-color mixing of blue from polymer matrix with orange yellow from silicate phosphor at an appropriate ratio.

A series of emission spectral profiles of composite polymer films with different weight percentage of silicate luminophor excited by the wavelength of 365 nm are shown in Fig. 8, and the dependency of the characteristic emission intensity at 420 nm and 580 nm on the content of silicate phosphor is illustrated in Fig. 9. Compared with that of pure polymer matrix film, the intensity of the emission peak at 420 nm of the composite films is stronger. Besides, the intensity of the characteristic emission peak at 580 nm monotonically increases up to the maximum value as the content is 0.5 wt%. This mutual enhancement is perhaps due to the restricted molecular chain motion of PS-PGMA when silicate phosphor is

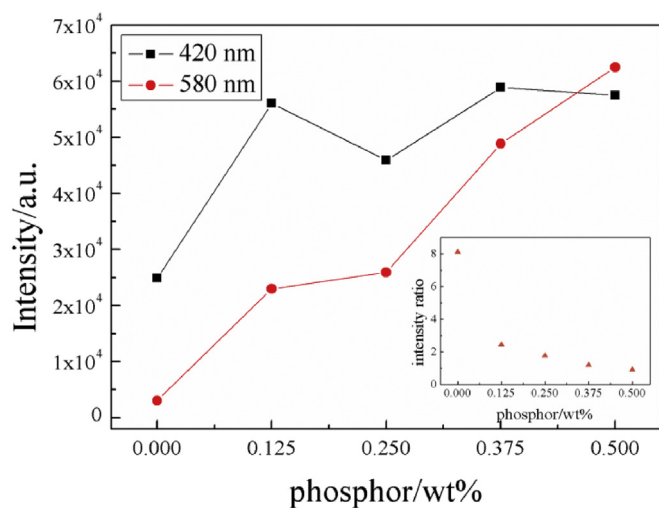


Fig. 9. Dependency of the characteristic emission (420 nm, 580 nm) intensity of composite material on the content of Silicate phosphor. The inset is intensity ratio of emission peak at 420 nm and 580 nm.

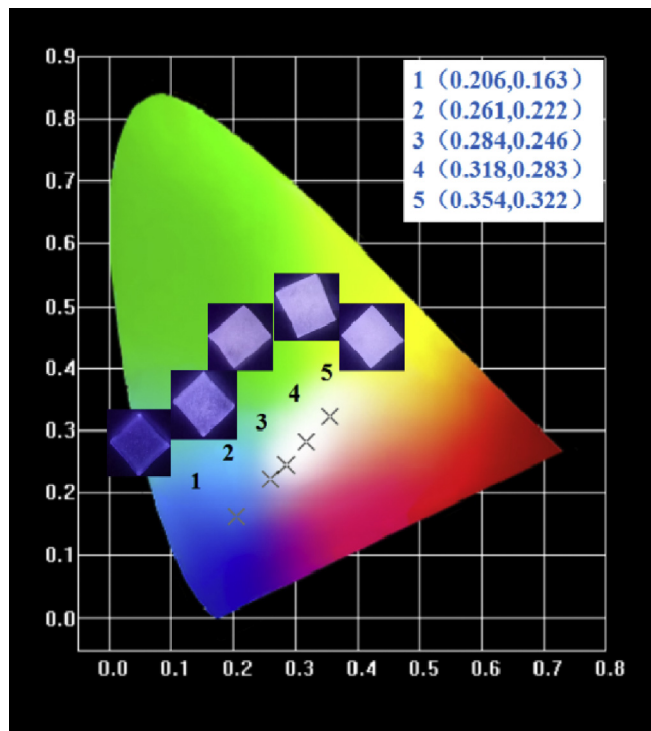


Fig. 10. The CIE chromaticity coordinates of the serial composite phosphors calculated from fluorescence emission spectra. The inset photographs are the polymeric white emitting film excited at 365 nm.

introduced. At the same time, the agglomeration of particles wrapped in polymer molecular chain is dramatically reduced (as can be confirmed by Fig. 3(c)) [23,24]. The inset profile in Fig. 9 is the intensity ratio of emission peak at 420 nm and 580 nm. When the content of silicate phosphor increases from 0 wt% to 0.5 wt%, the value of intensity ratio monotonically decreases from 7.21 to 0.93. The corresponding CIE chromaticity coordinates are calculated from the emission spectra and marked in Fig. 10. The coordinates change gradually from blue (numbered 1: 0 wt%) to white region (numbered 5: 0.5 wt%). And the optimal CIE coordinates (0.354, 0.322) of composite phosphor film locate in near white area. The inset photographs next to the corresponding numbers in Fig. 10 are the practical emission of composite phosphor films 1–5 under the irradiation with 365 nm. It is distinct that their gradually changing emission dovetails beautifully with the CIE chromaticity coordinates. Therefore, by varying the ratios of two ingredients, yellow-to-white PL color-tuning can be readily accomplished depending on the variation of the intensity ratios between the characteristic yellow emission and the blue luminescence. And white emission could be obtained, which comes from the simultaneous emission of blue emissions and comparable yellow emission.

3.3. Fluorescence lifetime

The decay curves for the luminescence of silicate phosphor, polymer matrix and composite film recorded at the excitation wavelength of 365 nm are shown in Fig. 11 black lines. The decay curves (red lines) were fitted into the exponential Eq. (1)

$$I(t) = A + B_1 \exp(-t/\tau_1) + B_2 \exp(-t/\tau_2) + B_3 \exp(-t/\tau_3) \quad (1)$$

where τ_1 , τ_2 and τ_3 are the lifetime parameters. B_1 , B_2 and B_3 are fitting constants. The average lifetime was calculated by Eq. (2)

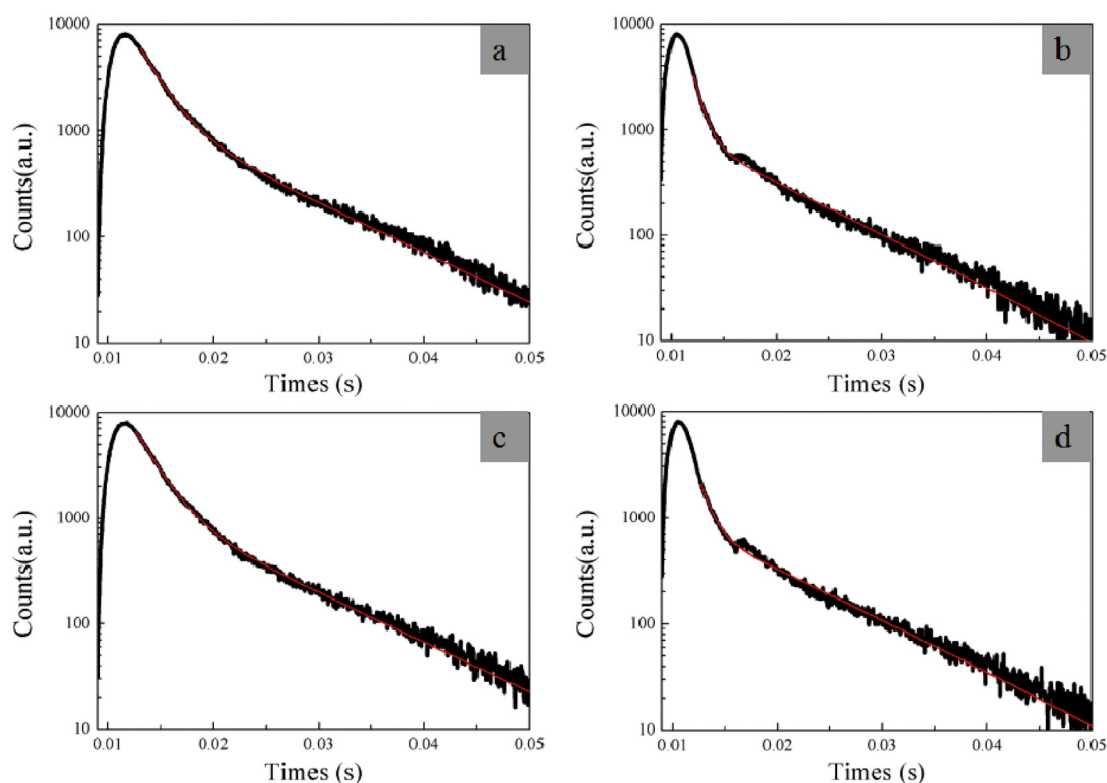


Fig. 11. The emission decay profiles and their corresponding fitting curves of (a) silicate phosphor ($\lambda_{\text{exc}} = 365$ nm, $\lambda_{\text{mon}} = 580$ nm), (b) polymer matrix film ($\lambda_{\text{exc}} = 365$ nm, $\lambda_{\text{mon}} = 420$ nm), (c) silicate phosphor embedded polymer film ($\lambda_{\text{exc}} = 365$ nm, $\lambda_{\text{mon}} = 580$ nm), (d) silicate phosphor embedded polymer film ($\lambda_{\text{exc}} = 365$ nm, $\lambda_{\text{mon}} = 420$ nm).

$$\langle \tau \rangle = \frac{B_1 \tau_1^2 + B_2 \tau_2^2 + B_3 \tau_3^2}{B_1 \tau_1 + B_2 \tau_2 + B_3 \tau_3} \quad (2)$$

All the parameters and calculated values are listed in Table 1.

It is noted that the lifetime of 5.7 μs for composite film is higher than that for raw silicate phosphor (5.3 μs) (emission wavelength is 580 nm). Similarly, the lifetime of 4.2 μs for composite film is higher than that of pure matrix film (2.9 μs) (emission wavelength is 420 nm). The result confirms the mutual enhancement of fluorescence for ingredients within composite film. The phenomenon is caused by the weakened interaction among silicate particles when silicate phosphors are embedded in co-polymer matrix [25]. In return, the movement and rearrangement of polymer chain segments are decreased owing to the analogous reinforced concrete frame structure. Ultimately, the probability of nonradiative transition is reduced.

3.4. WLED fabricated with silicate phosphor (0.5 wt%)/PS-PGMA composite

In order to further demonstrate the applicability as a

luminescent material for WLED device, silicate phosphor (0.5 wt %)/PS-PGMA composite material was encapsulated into 1 W lamp beads to generate white light. The brief process of encapsulation is as follows.

Fix a LED chip and spread the surface with a certain quantity of as-prepared composite polymer phosphor, followed by baking the LED covered with phosphor in an oven at 180 $^{\circ}\text{C}$ for 5 min. Then put it at room temperature for curing for 20 min. The optical material is solidly fixed on the surface of the NUV-LED chip by virtue of the content of epoxy groups.

Fig. 12 shows the luminance-voltage characteristics of the LED beads fabricated with composite phosphor and original LED with 365 nm chip. It is clearly seen that the maximum luminance (7980 cd/m^2) of the fabricated WLEDs is 2.6 times that (3075 cd/m^2) of original LED chip. Fig. 13 is the EL spectra of the LED beads when the driving voltage is 3.2 V. The spectrum of the fabricated WLED exhibits blue and yellow emissions, which are admirably consistent with those of the photoluminescent spectrum (see Fig. 8). What calls for special attention is that the maximum of blue peak has a red shift of 11 nm. This is probably caused by the formation of internal stress during the solidification process on the surface of the chip. The corresponding CIE chromaticity coordinates

Table 1

Parameters and calculated values of decay curves for the luminescence of silicate phosphor, polymer matrix and composite film.

Material number	Emission wavelength (nm)	B ₁	B ₂	B ₃	τ_1 (ns)	τ_2 (ns)	τ_3 (ns)	T (μs)
a	580	4725	1287	/	2145	8316	/	5.3
b	420	2587	781	−106	933	8852	2838509	2.9
c	580	−571	5856	1297	313	2110	9247	5.7
d	420	1413	736	−7	1052	9165	65875	4.2

Note: Material number **a** and **b** in Table 1 respectively represent silicate phosphor and polymer matrix film, **c** and **d** represent silicate phosphor composite polymer film.

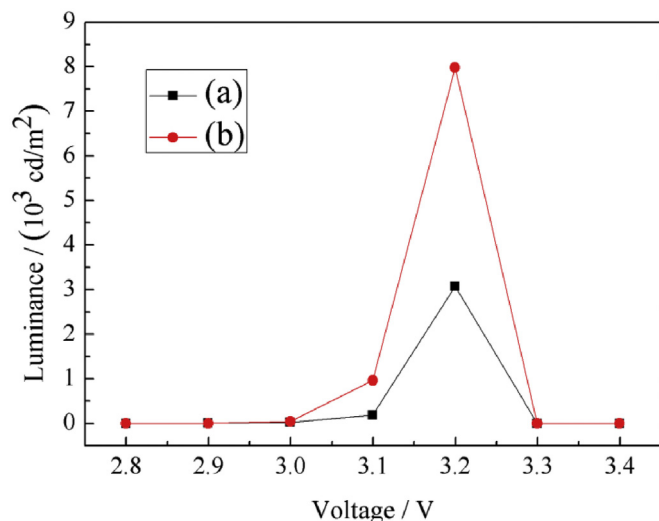


Fig. 12. The luminance-voltage curves of the LEDs fabricated with silicate phosphor (0.5 wt%)/PS-PGMA composite powder and the original LED (365 nm UV chip) without phosphor (b).

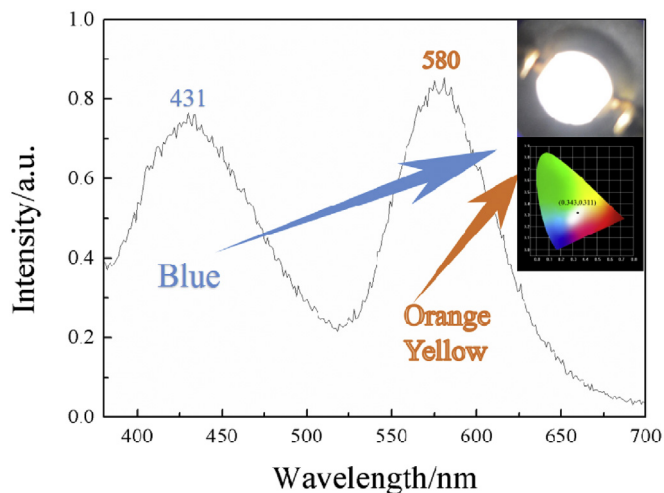


Fig. 13. The electroluminescence (EL) of the LEDs fabricated with silicate phosphor (0.5 wt%)/PS-PGMA composite powder. The inset photo shows visible emissions from LED lamp beads (top) and the corresponding CIE chromaticity coordinate (below).

based on the EL spectra (shown in the inset photographs) are (0.343, 0.311), presenting a small deviation from (0.354, 0.322) calculated from photoluminescence spectra. The negligible deviation of chromaticity coordinates according to EL spectra and PL spectra suggests the stability of the phosphor in its luminous performance when fabricated into white LEDs.

Moreover, the fabricated WLED has an applicable color temperature (5917 K) and extremely high color rendering index (84.6) compared with WLEDs based on blue InGaN chip with YAG:Ce³⁺ or other inorganic phosphors [26,27]. This is due to more red ingredient in the luminescence spectrum of silicate phosphor, which is conducive to higher CRI value in the white LEDs. Fig. 14 is the digital photograph of the practical emission of transparent silicate phosphor doped composite fluorescent film over COB LED (light emitting area is 25 mm × 25 mm). Obviously, this kind of composite phosphor with the ease in fabrication of large size films can satisfy the illumination requirements as a promising optical material for application in lighting area.

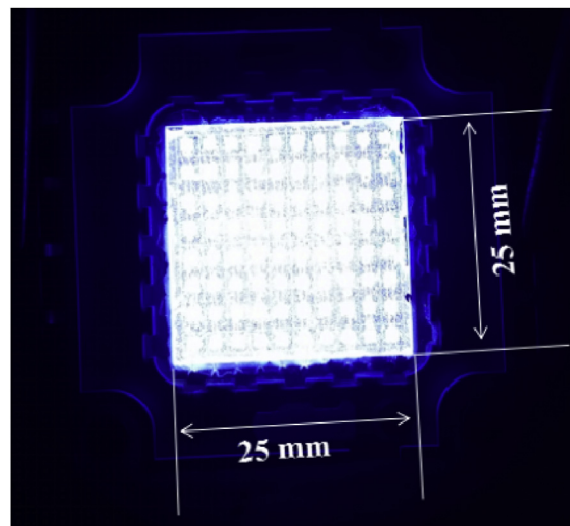


Fig. 14. Photograph of transparent silicate phosphor doped composite fluorescent film over COB LEDs, inset shows a collection of colorful candies illuminated with the same device.

4. Conclusions

In conclusion, a silicate phosphor-containing transparent composite film was prepared through a readily accessible melt compounding process, which is able to emit white light on the base of a blue emitting co-polymer matrix PS-PGMA under excitation of 365 nm. By varying the ratios of silicate phosphor and PS-co-PGMA, yellow-to-white PL color-tuning can be readily accomplished depending on the variation of the intensity ratios between the characteristic luminescence. The strongly blue emission from the PS-PGMA matrix is sufficient to balance the yellow emission of silicate phosphor to generate white-light (intensity ratios value is 0.93). Simple NUV-based WLED was fabricated using the silicate phosphor doped composite material, showing the maximum brightness of 7980 cd/m², T_c of 5917 K, CRI of 84.6. When a larger size composite film was simply assembled with COB, large area white lighting device with a certain color reduction came into being. All the results indicate that this kind of silicate-phosphor containing transparent composite film has practical significance for the packaging of remote phosphor converted white LEDs.

Acknowledgements

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